

MANTHAN PENDKE

AI/ML Engineer | Applied Machine Learning & LLM Systems

mpendke9@gmail.com | +91 8482916576 | [Portfolio](#) | [LinkedIn](#)

SUMMARY

AI/ML engineer with hands-on experience building predictive maintenance, computer vision, and LLM-based systems — including RAG pipelines, LLaMA-based models, and imbalanced-data handling with SMOTE. Published IEEE researcher with end-to-end experience across data preprocessing, model training/fine-tuning, evaluation, and deployment, plus the full-stack skills to ship ML-powered dashboards.

TECHNICAL SKILLS

AI/ML: TensorFlow, PyTorch, OpenCV, Computer Vision, NLP, RAG, LLaMA, LangChain, FAISS, Vector Databases, SMOTE

ML Workflow: Model fine-tuning, classification, root-cause analysis, recommendation systems, MLOps (deployment & monitoring)

Languages: Python, Core Java, JavaScript, C++

Backend/Frontend: React.js, Node.js, Express.js, REST APIs, HTML5, CSS3

Databases: MongoDB, SQL Server, MySQL

Cloud & Tools: AWS, Git, GitHub

EXPERIENCE

AI Intern — Expleo

July 2025 – Aug 2025

- Developed a predictive maintenance system for CNC machines using Machine Learning and LLaMA-based LLMs to forecast alarms and minimize downtime.
- Implemented Retrieval-Augmented Generation (RAG) and SMOTE to enhance alarm diagnostics and improve model performance on imbalanced datasets.
- Built and fine-tuned ML models for alarm classification and root-cause analysis using historical sensor and operational data.
- Designed a React.js dashboard for real-time alarm visualization, integrating predictive analytics directly into the CNC machine workflow.

Developer Intern — Endava Solutions

Jan 2026 – May 2026

- Designed a relational database schema in SQL Server, applying ER modeling and normalization — directly transferable to structuring training data and feature stores.
- Built backend modules in Java using OOP principles for data processing and business logic implementation.
- Developed a personalized recommendation system using content-based and collaborative filtering (both user-based and item-based) along with matrix factorization methods to generate tailored recommendations, applying core AI/ML techniques.
- Performed functional, database, and UI testing to validate system reliability and data integrity.

PROJECTS

Disaster Information Aggregation Software

- Developed a multimodal disaster classification system using deep learning to identify disaster types and assess severity from images and unstructured data.
- Implemented computer vision pipelines — image preprocessing, augmentation, OCR, and object detection — to improve model performance.
- Trained and evaluated deep learning models on large-scale disaster datasets for accurate classification and severity prediction.
- Integrated visual, textual, and geospatial data into an end-to-end real-time disaster intelligence and alerting system.

- Published results as an IEEE research paper covering the multimodal NLP, computer vision, and ML architecture.

College Prediction System

- Developed a machine learning-based admission prediction system to recommend eligible colleges, branches, and admission rounds from student rank and category.
- Designed custom prediction algorithms using historical admission data for accurate, personalized recommendations.
- Automated data preprocessing, model training, and prediction workflows to improve scalability and reduce manual effort.
- Applied MLOps practices for model deployment, monitoring, and continuous performance improvement.

Location-Optimized Web System for Jobs

- Built a recruitment platform with location-based search and proximity-driven recommendation logic.
- Implemented secure authentication and real-time tracking for the application workflow.

CERTIFICATIONS

AWS Certified Cloud Practitioner — Amazon Web Services (May 2026)

Introduction to Large Language Models (LLMs) — Udemy (April 2025)

EDUCATION

B.Tech, Electronics & Telecommunication Engineering — 8.89 CGPA

Nov 2022 – June 2026

HSC — 74.17%

2021 – 2022

ACHIEVEMENTS

Published an IEEE research paper on Disaster Information Aggregation using Multimodal Artificial Intelligence, integrating NLP, Computer Vision, and Machine Learning techniques.